

MDGen: Generative Molecular Dynamics Simulation

A Comprehensive Technical Survey with Background,
Prior Work, Architecture, and Analysis

Ditsa Mallick

School of Computing and Augmented Intelligence
Arizona State University, Tempe AZ 85281
dmallic1@asu.edu

Based on:

Jing, B., Stärk, H., Jaakkola, T., & Berger, B. (2024).
Generative Molecular Dynamics. arXiv:2409.17808. NeurIPS 2024.

Abstract

MDGen is the first generative model to produce temporally coherent, full-backbone protein trajectories at $\sim 10,000\times$ the speed of explicit-solvent molecular dynamics (MD). This survey provides a comprehensive technical exposition covering protein biophysics, denoising diffusion probabilistic models on non-Euclidean manifolds ($SO(3)$, T^4), $SE(3)$ -equivariance, and the full landscape of prior work—Normal Mode Analysis, Boltzmann generators, and independent-frame diffusion models. The MDGen architecture alternates $SE(3)$ -equivariant Invariant Point Attention (spatial) with temporal axial attention (kinetic), enabling joint modelling of equilibrium thermodynamics and kinetic temporal structure. Trained on the ATLAS database (1,400 proteins $\times$ 300 ns each), MDGen achieves RMSF Pearson $r \approx 0.87$, Ramachandran JSD ≈ 0.08 , and correctly reproduces autocorrelation functions and frame-to-frame step-size distributions—metrics no prior independent sampler can reproduce.

1 Introduction

Proteins are the primary molecular machines of living systems. They catalyse chemical reactions, transmit cellular signals, provide structural integrity, and execute the vast majority of biochemical functions essential for life. A protein's biological activity depends critically not only on its average three-dimensional structure but on its dynamical behaviour: the ensemble of conformations it explores at physiological temperature, the timescales of interconversion between states, and the kinetic pathways connecting them (Adcock & McCammon, 2006).

Despite decades of progress in experimental structure determination—X-ray crystallography, cryo-EM, NMR spectroscopy—these methods report static or ensemble-averaged snapshots. Molecular dynamics (MD) simulation fills this gap by numerically integrating Newton's equations of motion for every atom, generating detailed time-series of atomic positions. Modern GPU-accelerated MD packages can simulate a 100-residue protein at ~ 100 – 200 ns per day. Yet biologically important timescales—domain motions, loop gating, folding transitions—occur on microsecond to millisecond timescales, spanning 9–12 orders of magnitude above the obligatory 2 fs integration time step. This timescale gap makes MD prohibitively expensive for large-scale ensemble characterisation or drug discovery campaigns.

Core motivation. MDGen asks: can a neural network, trained on a large corpus of MD trajectories, learn to generate new trajectory segments that faithfully reproduce both the equilibrium thermodynamics AND the kinetic temporal structure—without running the simulation itself?

All prior generative models for protein conformations operate in an i.i.d. sampling regime: each sample is drawn independently from an approximation of the equilibrium distribution. Such models cannot reproduce temporal autocorrelation functions, kinetically realistic step sizes between consecutive frames, or any observable that depends on the ordering of configurations in time. MDGen (Jing et al., NeurIPS 2024) addresses this gap by posing the generative problem at the level of entire trajectory segments: given a protein sequence, generate T consecutive backbone frames whose joint distribution matches that of a real MD simulation. The key technical contribution is an architecture alternating SE(3)-equivariant Invariant Point Attention (IPA) with novel temporal axial attention. Trained on ATLAS (1,400 proteins $\times$ 300 ns each), MDGen achieves RMSF Pearson $r \approx 0.87$ and correct temporal kinetics at $\sim 10,000\times$ the speed of equivalent MD.

2 Biological and Physical Background

2.1 Proteins: Structure and Function

A protein is a linear polymer of amino acid residues connected by peptide bonds. Twenty standard amino acids, differing only in their side chain (the group branching from the

alpha-carbon, C α), are encoded by the genetic code. Proteins range from tens to thousands of residues in length; a typical globular protein has 100–500 residues.

The structural hierarchy of proteins is decomposed into four levels:

- **Primary structure:** amino-acid sequence, a string over a 20-letter alphabet.
- **Secondary structure:** backbone regularities— α -helices (right-handed coils stabilised by i→i+4 hydrogen bonds) and β -strands (extended chains forming hydrogen-bonded β -sheets).
- **Tertiary structure:** complete 3D fold of a single polypeptide chain, governed by hydrophobic packing, electrostatics, hydrogen bonding, and van der Waals interactions.
- **Quaternary structure:** assembly of multiple polypeptide chains into a complex.

The protein backbone consists of the repeating N-C α -C=O unit running through every residue. Backbone conformation is almost completely determined by two dihedral angles: ϕ (N-C α bond rotation) and ψ (C α -C bond rotation). MDGen operates exclusively on backbone heavy atoms, representing each residue as a rigid frame, with side chains encoded separately as torsion angles χ_1 – χ_4 .

2.2 The Protein Energy Landscape and Conformational Ensembles

A protein in solution at temperature T does not adopt a single static conformation. Instead, it continuously fluctuates according to Boltzmann statistics. The probability of finding the system in conformation x is given by the canonical ensemble distribution:

$$p(x) \propto \exp\left(-\frac{E(x)}{k_B T}\right) \tag{1}$$

where E(x) is the potential energy, k_B is Boltzmann's constant, and k_BT $\approx$ 0.6 kcal/mol at 300 K. The energy landscape of a protein (Frauenfelder et al., 1991) is a high-dimensional surface in conformation space. Key features relevant to MDGen:

- Multiple minima: most functional proteins populate more than one conformation (e.g. open/closed enzyme active sites).
- Hierarchical roughness: fast local motions (10 fs) nested within slower large-scale rearrangements (μ s–ms).
- Anharmonicity: large-amplitude motions cannot be described by a harmonic approximation—a fundamental limitation of Normal Mode Analysis.

The biologically relevant object is the conformational ensemble—the full distribution p(x)—not a single structure. Dynamical properties such as conformational change rates, domain motion timescales, and cryptic binding pocket populations all depend on the shape of this distribution and the barriers between its modes.

2.3 Molecular Dynamics Simulation

MD simulation generates samples from the conformational ensemble by numerically integrating Newton's equations of motion:

$$m_i \frac{d^2 \mathbf{r}_i}{dt^2} = \mathbf{F}_i = -\nabla_{\mathbf{r}_i} E(\mathbf{r}_1, \dots, \mathbf{r}_N) \quad (2)$$

The empirical force field E comprises: (i) bonded terms—harmonic bond stretches, angle bends, and periodic dihedral torsions calibrated to quantum chemical calculations; (ii) non-bonded terms—Lennard-Jones 6-12 van der Waals potential and Coulomb electrostatics (typically treated via Particle Mesh Ewald for long-range contributions). Widely used force fields include AMBER ff14SB (the force field used for ATLAS training data), CHARMM36m, and OPLS-AA.

The timescale problem. The fastest atomic vibrations (~ 10 fs) impose an integration time step ceiling of $\Delta t = 2$ fs. Observing biologically important motions requires 10^8 - 10^{11} steps (100 ns-100 μ s). A single microsecond simulation of a 100-residue protein on a modern A100 GPU cluster requires ~ 1 -2 weeks wall-clock time. Screening thousands of variants—as required in drug discovery—is practically infeasible.

2.4 Trajectory Representation for MDGen

An MD trajectory is a time-ordered sequence of frames $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T)$, where each frame $\mathbf{x}_t$ specifies the full atomic coordinates at time t . In MDGen, the relevant degrees of freedom per residue i at time t are a rigid body frame in $SE(3)$ plus torsion angles on the 4-torus:

$$\mathbf{x} = \{(R_i, \mathbf{t}_i, \chi_i)\}_{i=1}^N \in SE(3)^N \times \mathbb{T}^{4N} \quad (3)$$

A trajectory window of T frames is the joint object $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T)$ over which MDGen's diffusion process is defined. Training uses $T = 16$ frames, corresponding to approximately 16 ns at the ATLAS 1 ns frame rate.

3 Theoretical Background: Diffusion Models

3.1 Denoising Diffusion Probabilistic Models (DDPMs)

Denoising diffusion probabilistic models (DDPMs; Ho et al., 2020) define a generative model through two coupled Markov chains: a forward process that gradually destroys structure by adding Gaussian noise, and a reverse process learned to reconstruct data from noise.

Forward process. Given a clean sample $\mathbf{x}_0 \sim q_{\text{data}}(\mathbf{x})$, the forward process adds Gaussian noise over S discrete steps:

$$q(\mathbf{x}_s | \mathbf{x}_{s-1}) = \mathcal{N}(\mathbf{x}_s; \sqrt{1 - \beta_s} \mathbf{x}_{s-1}, \beta_s \mathbf{I}) \quad (4)$$

where $\beta_s \in (0,1)$ is a pre-specified noise schedule. The marginal at step s has a closed form with $\bar{\alpha}_s = \prod_{k=1}^s (1 - \beta_k)$:

$$q(\mathbf{x}_s | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_s; \sqrt{\bar{\alpha}_s} \mathbf{x}_0, (1 - \bar{\alpha}_s) \mathbf{I}) \quad (5)$$

As $s \rightarrow S$, $\bar{\alpha}_s \rightarrow 0$ and the distribution approaches $N(0, I)$ — pure isotropic noise. The training objective minimises noise prediction error:

$$\mathcal{L} = \mathbb{E}_{S, \mathbf{x}_0, \varepsilon} [\|\varepsilon - \varepsilon_\theta(\sqrt{\bar{\alpha}_s} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_s} \varepsilon, s)\|^2] \quad (6)$$

3.2 Score-Based Models and the Continuous-Time SDE Perspective

Song et al. (2021) cast diffusion models as continuous-time stochastic differential equations (SDEs), providing a unified theoretical framework. The forward SDE is:

$$d\mathbf{x} = f(\mathbf{x}, t) dt + g(t) d\mathbf{W} \quad (7)$$

where $\mathbf{W}$ is a standard Brownian motion. The reverse SDE (Anderson, 1982) runs backwards from noise to data:

$$d\mathbf{x} = [f(\mathbf{x}, t) - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})] dt + g(t) d\tilde{\mathbf{W}} \quad (8)$$

The key quantity is the score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$, which points toward regions of higher probability density. A score network $s_\theta(\mathbf{x}, t) \approx \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ is trained via denoising score matching (Vincent, 2011). DDIM (Song et al., 2021) provides a deterministic ODE equivalent enabling high-quality sampling in ~ 50 steps instead of 500:

$$\mathbf{x}_{s-1} = \sqrt{\bar{\alpha}_{s-1}} \left(\frac{\mathbf{x}_s - \sqrt{1 - \bar{\alpha}_s} \varepsilon_\theta}{\sqrt{\bar{\alpha}_s}} \right) + \sqrt{1 - \bar{\alpha}_{s-1}} \varepsilon_\theta \quad (9)$$

3.3 Diffusion on Non-Euclidean Manifolds

Standard Gaussian diffusion assumes data lies in flat Euclidean space $\mathbb{R}^d$. Protein backbone frames involve two non-Euclidean components:

- Rotations $R_i \in \text{SO}(3)$: a compact 3D Lie group with non-trivial curved geometry.
- Torsion angles $\chi_j \in S^1$ (the circle), with period 2π ; side-chain state lives on the 4-torus $T^4 = (S^1)^4$.

Diffusion on $\text{SO}(3)$: MDGen uses the Isotropic Gaussian on $\text{SO}(3)$ (IGSO3; De Bortoli et al., 2022) distribution:

$$q(R_s | R_0) = \text{IGSO3}(R_s; R_0, \sigma_s^2) \quad (10)$$

whose score uses the Riemannian gradient—the analogue of $\nabla_{\mathbf{x}} \log p$ on the curved $\text{SO}(3)$ manifold, computed via the heat kernel. For torsion angles $\theta \in (-\pi, \pi]$, the correct diffusion distribution is a wrapped Gaussian (Jing et al., 2022):

$$q(\theta_s | \theta_0) = \sum_{k=-\infty}^{\infty} \mathcal{N}(\theta_s; \theta_0 + 2\pi k, \sigma_s^2) \quad (11)$$

3.4 SE(3) Equivariance

A fundamental symmetry of physical systems in 3D is Euclidean equivariance. The group $\text{SE}(3) = \text{SO}(3) \ltimes \mathbb{R}^3$ consists of all rotations and translations of $\mathbb{R}^3$. A map f is

SE(3)-equivariant if:

$$f(g \cdot \mathbf{x}) = \rho_{\text{out}}(g) f(\mathbf{x}), \quad g \in SE(3) \quad (12)$$

For coordinate outputs, rotating the input rotates the output by the same rotation.

SE(3)-equivariant networks (Fuchs et al., 2020; Satorras et al., 2021) architecturally enforce this constraint exactly. This is essential: the physics of a protein does not change if we rotate or translate the entire coordinate system. A model that violates this symmetry must learn it from data, wasting capacity and risking generalisation failure.

3.5 Backbone Frame Representation

The AlphaFold2 structure module (Jumper et al., 2021) introduced the backbone frame representation that MDGen inherits. Each residue i is assigned a rigid body in SE(3):

- Translation $\mathbf{t}_i \in \mathbb{R}^3$: the position of the C α atom.
- Rotation $\mathbf{R}_i \in \text{SO}(3)$: a local coordinate frame defined by the N, C α , and C atoms, encoding backbone orientation.

Together, $\mathbf{T}_i = (\mathbf{R}_i, \mathbf{t}_i)$ defines a frame in which the local chemical environment of residue i appears in a standardised orientation. The full protein backbone is represented as N frames $\{\mathbf{T}_i\}_{i=1}^N \in \text{SE}(3)^N$.

4 Prior Work and Its Limitations

Understanding MDGen's novelty requires surveying the space of prior methods for computational characterisation of protein conformational dynamics. We organise this survey into four categories.

4.1 Normal Mode Analysis (NMA)

Normal Mode Analysis (Tirion, 1996; Brooks & Karplus, 1983) approximates the protein energy landscape as a multidimensional quadratic (harmonic) bowl centred on a known minimised structure. Under this approximation, the equations of motion separate into $3N-6$ independent harmonic oscillators (normal modes), each with a characteristic frequency. The lowest-frequency modes correspond to the largest-scale, most collective motions and can predict B-factors and RMSF profiles.

Strengths of NMA:

- Computationally cheap: a single eigendecomposition of the Hessian matrix.
- Provides physically interpretable modes (e.g. a 'breathing' mode of a capsid).
- Reasonable agreement with MD-derived RMSF for well-ordered secondary structure.

Fundamental limitations:

- Harmonic approximation: NMA is strictly valid only near the energy minimum. Loop motions, domain swings, and any transition requiring barrier crossing are outside the scope of the harmonic model.
- No kinetics: NMA produces independent samples from a Gaussian approximation to the

equilibrium ensemble; it has no notion of temporal ordering or transition rates.

- Single minimum: NMA cannot represent multi-modal distributions or conformational heterogeneity between distinct basins.
- In direct comparison on the ATLAS test set, NMA achieves RMSF Pearson $r \approx 0.64$ — substantially below MDGen's $r \approx 0.87$.

4.2 Boltzmann Generators and Flow-Based Equilibrium Samplers

Boltzmann generators (Noé et al., 2019) use normalising flows (bijective, differentiable maps) to learn a direct mapping between a tractable Gaussian base distribution and the Boltzmann distribution $p(x) \propto \exp(-E(x)/k_{BT})$. Samples can be drawn at test time without simulating the dynamics.

Strengths:

- Principled probabilistic model with exact likelihood.
- Can provide reweighting factors to correct for force-field imperfections.

Fundamental limitations:

- Scalability: normalising flows for all-atom protein configurations of 100–500 residues remain computationally intractable; most demonstrations are restricted to ≤ 50 -residue systems.
- No temporal coherence: like NMA, Boltzmann generators produce i.i.d. equilibrium samples. There is no mechanism for temporally ordered trajectories.
- Requires the energy function: training typically requires evaluating $E(x)$, coupling the method to the expensive force field.

4.3 Independent-Frame Protein Diffusion Models

A family of recent diffusion models generates protein structures by learning to reverse a noising process on backbone frames, producing independent equilibrium samples.

- **FrameDiff**: et al., ICML 2023): diffuses over $SE(3)^N$ backbone frames; samples drawn independently to form a structural ensemble.
- **EigenFold**: et al., 2023): decomposes protein structure along a hierarchy of spring-network eigenmodes, performing diffusion along each independently.
- **AlphaFlow**: et al., ICML 2024): fine-tunes AlphaFold2 as a flow-matching model conditioned on MSAs, enabling ensemble prediction for evolutionary context.
- **BioEFlow**: et al., Science 2025): large-scale diffusion model on MD trajectories and experimental data; strong equilibrium ensemble predictions.

Shared limitation: temporal incoherence. All of the above generate INDEPENDENT conformational samples. While they may approximate the marginal equilibrium distribution $p(x)$, they cannot reproduce ANY temporal property: (1) the autocorrelation function $C(\tau)$ drops to zero immediately at lag $\tau \geq 1$ for i.i.d. samplers; (2) frame-to-frame step size is arbitrarily large instead of $\sim 0.1\text{--}0.3 \text{ \AA}$; (3) transition pathway information is completely inaccessible.

4.4 Static Structure Prediction Models

- **AlphaFold2** (Jumper et al., 2021): near-experimental accuracy for single lowest-energy structure prediction. Produces one structure per input; not designed for ensemble generation or dynamics.
- **ESMFold** (Baerentzen et al., 2023): extends ESM-2 language model to fast single-structure prediction without MSA. MDGen uses it only for initial conditioning structure.
- **RFdiffusion** (Chaitin et al., 2023) and **Chroma** (Ingraham et al., 2023): diffusion models for protein design—generating novel backbone conformations satisfying functional constraints. Not evaluated on MD ensemble fidelity metrics.

4.5 Summary Comparison Table

Method	Equil.	Ensemble	Temporal Coh.	Speed vs MD	Large Proteins
MD simulation	[Y]	[Y]		1 × (baseline)	[Y]
NMA	~	[N]		~1000×	[Y]
Boltzmann generators	[Y]	[N]		~1000×	[N]
EigenFold / AlphaFlow	[Y]	[N]		~10000×	[Y]
BioEmu	[Y]	[N]		~10000×	[Y]
MDGen (this work)	[Y]	[Y]		~10000×	[Y]

5 MDGen: Method

5.1 Problem Formulation

Definition (Trajectory generation). Given a protein amino-acid sequence $s = (s_1, \dots, s_N)$, generate a trajectory window $X = (x_1, \dots, x_T) \in (\text{SE}(3)^N \times T^{\{4N\}})^T$ such that the joint distribution $p_\theta(X|s)$ approximates the distribution over T-frame windows drawn from equilibrium MD trajectories.

This formulation imposes two simultaneous requirements:

- Equilibrium marginal correctness: for any single time t , the marginal $p(x_t|s)$ matches the MD equilibrium distribution.
- Temporal coherence: the joint distribution $p(x_t, x_{t+1}|s)$ reflects the physically correct Markov transition structure of MD dynamics.

5.2 Forward Diffusion Process over Trajectory Space

MDGen applies independent noise to each trajectory frame, keeping the forward process factorised over the time axis. For a clean trajectory $X^{(0)}$ at diffusion level $s=0$ and noisy trajectory $X^{(s)}$ at level $s \in [0,S]$:

$$q(\mathbf{X}^{(s)}|\mathbf{X}^{(0)}) = \prod_{t=0}^T q(\mathbf{x}_t^{(s)}|\mathbf{x}_t^{(0)}) \quad (13)$$

Within each frame, diffusion is factorised over geometric components:

$$q(\mathbf{x}_t^{(s)}|\mathbf{x}_t^{(0)}) = q_{\mathbb{R}^3}(\mathbf{t}^{(s)}|\mathbf{t}^{(0)}) \cdot q_{SO(3)}(R^{(s)}|R^{(0)}) \cdot q_{\mathbb{T}}(\chi^{(s)}|\chi^{(0)}) \quad (14)$$

Key insight: although frames are noised independently in the forward process, the reverse process denoises ALL T frames jointly. Temporal correlations are entirely a property of the LEARNED denoising network—encoded by the temporal attention layers.

5.3 Neural Architecture

5.3.1 Sequence Encoder: ESM-2

The protein sequence s is embedded by the pre-trained protein language model ESM-2 (Lin et al., 2023) into per-residue feature vectors $\mathbf{h}_i^{\text{seq}} \in \mathbb{R}^{d_{\text{seq}}}$. ESM-2 is kept frozen during MDGen training. Its representations capture evolutionary and physicochemical information about each residue, serving as a rich conditioning signal projected to internal model dimension d via a learned linear layer.

5.3.2 Spatial Attention: Invariant Point Attention (IPA)

At each trajectory frame t , inter-residue information exchange is mediated by Invariant Point Attention (IPA; Jumper et al., 2021). IPA augments standard multi-head attention with 3D query/key point pairs expressed in each residue's local backbone frame. The attention score between residues i and j is:

$$a_{ij} = \text{softmax}_j \left(\frac{\mathbf{q}_i^T \mathbf{k}_j}{\sqrt{d}} - \frac{\gamma}{2} \sum_h \|T_i \mathbf{p}_i^{(h)} - T_j \mathbf{p}_j^{(h)}\|^2 \right) \quad (15)$$

where $T_i = (R_i, \mathbf{t}_i)$ maps local frame points to ^{h} global coordinates, $\mathbf{p}_i^{(h)}$ are learned 3D query points in residue i 's local frame, and $\gamma > 0$ is a learnable scalar. The attention is SE(3)-invariant: $\|T_i \mathbf{p} - T_j \mathbf{q}\|$ is unchanged by any global rotation or translation. Two residues that are distant in sequence but close in 3D space receive a high attention weight, enabling the model to reason about spatial contacts across the entire structure.

5.3.3 Temporal Attention: Axial Attention Across Time (Key Innovation)

The defining architectural innovation in MDGen is the introduction of axial temporal attention (Ho et al., 2019). After the spatial IPA pass (across residues at fixed t), a temporal attention pass operates across all T time steps for each residue i :

$$\mathbf{h}_{i,t}^{\text{new}} = \sum_{t'}^T \alpha_{tt'} \mathbf{v}_{i,t'}, \quad \alpha_{tt'} = \text{softmax}_{t'} \left(\frac{\mathbf{q}_{i,t} \cdot \mathbf{k}_{i,t'}}{\sqrt{d}} \right) \quad (16)$$

The temporal attention layer allows every time step to attend to every other time step within the trajectory window. This is precisely the mechanism through which MDGen learns that consecutive frames should be physically similar: the loss penalises temporally incoherent predictions, and temporal attention weights adapt to propagate continuity

information along the time axis.

The spatial and temporal attention layers alternate in a stack of L blocks, constituting an axial attention transformer over the $N \times T$ grid of (residue, frame) tokens:

- IPA_l: spatial pass—all N residues exchange information at each fixed frame t.
- TempAttn_l: temporal pass—each residue i attends across frames $t=1, \dots, T$.
- FFN_l: position-wise feed-forward network.

5.3.4 Score Head

After L alternating blocks, separate prediction heads convert the latent representation $h_{\{i,t\}}^{(L)}$ into predicted noise for each geometric component: $\hat{\epsilon}^{\text{trans}}_{\{i,t\}} \in \mathbb{R}^3$, $\hat{\epsilon}^{\text{rot}}_{\{i,t\}} \in \text{so}(3)$, $\hat{\epsilon}^{\text{tors}}_{\{i,t\}} \in \mathbb{R}^4$. The rotation score is expressed as a vector in the tangent space of the Lie algebra $\text{so}(3) \sim \mathbb{R}^3$. Total parameter count: $\sim 100\text{M}$.

5.4 Training Objective

The training loss is a weighted sum of denoising score matching losses for each geometric component, plus a structural violation penalty:

$$\mathcal{L} = \mathbb{E}_{S, \mathbf{X}^{(0)}, \mathbf{X}^{(s)}} [\lambda_{\text{trans}} \mathcal{L}_{\mathbb{R}^3} + \lambda_{\text{rot}} \mathcal{L}_{\text{SO}(3)} + \lambda_{\text{tors}} \mathcal{L}_{\mathcal{T}} + \lambda_{\text{viol}} \mathcal{L}_{\text{viol}}] \quad (17)$$

- $\mathcal{L}_{\text{trans}} = (1/NT) \sum_{\{i,t\}} \|\hat{\epsilon}^{\text{trans}} - \epsilon^{\text{trans}}\|^2$: standard MSE on C α displacements. ($\lambda=1.0$)
- $\mathcal{L}_{\text{rot}}$: Riemannian score matching loss on the SO(3) manifold. ($\lambda \approx 1.0$)
- $\mathcal{L}_{\text{tors}}$: wrapped Gaussian score matching for torsion angles. ($\lambda \approx 0.5$)
- $\mathcal{L}_{\text{viol}}$: pairwise steric clash penalty + backbone bond-geometry violations. ($\lambda \approx 0.01$)

All N residues and all T=16 frames contribute to each gradient update. Diffusion noise level $s \sim \text{Uniform}(0, S)$ with $S=500$. Optimiser: AdamW, lr=1e-4, cosine annealing, batch size ≈ 64 windows, on $8 \times \text{A100 GPUs}$.

5.5 Inference: Sampling

Sampling runs the reverse diffusion starting from pure noise for all T frames simultaneously: translations $\sim \text{N}(0, I)$, rotations $\sim \text{Uniform}(\text{SO}(3))$, torsions $\sim \text{Uniform}(\mathcal{T}^4)$. MDGen supports two modes:

- DDPM (500 steps): stochastic reverse process; highest sample quality; $\sim$ minutes per trajectory.
- DDIM (50 steps): deterministic ODE; $10\times$ faster than DDPM with minimal quality loss; $\sim$ seconds per trajectory.

Both modes produce trajectories approximately $10,000\times$ faster than equivalent explicit-solvent MD simulation.

5.6 Trajectory Extension via Diffusion Inpainting

MDGen can generate trajectories conditioned on a known conformation $\mathbf{x}_1^{(0)}$ (e.g. a crystal structure) using diffusion inpainting (Lugmayr et al., 2022). During reverse diffusion, the first-frame variable is periodically replaced with its correctly-noised counterpart:

$$\mathbf{x}_1^{(s)} \leftarrow \sqrt{\tilde{\alpha}_s} \mathbf{x}_1^{(0)} + \sqrt{1 - \tilde{\alpha}_s} \varepsilon \quad (18)$$

Interpolation mode: conditioning on BOTH $\mathbf{x}_1^{(0)}$ and $\mathbf{x}_T^{(0)}$ generates a plausible transition pathway between two known protein conformations (e.g. open $\leftrightarrow$ closed states of adenylate kinase). This capability for transition path sampling without enhanced-sampling MD has no equivalent in any prior independent-frame sampler.

6 Training Data: The ATLAS Database

MDGen is trained on the ATLAS (Atlas of Trajectory Landscapes and Associated Simulations) database (Vander Meersche et al., Nucleic Acids Research, 2024), the largest open repository of protein MD trajectories assembled specifically to support machine learning on protein dynamics.

Simulation protocol:

- Force field: AMBER ff14SB with explicit TIP3P water, NPT ensemble, 300 K, 1 atm.
- Three independent 100 ns replicas per protein $\rightarrow$ 300 ns total per protein.
- Frames saved every 1 ns $\rightarrow$ $\sim$ 300 frames per protein.
- Simulation time step: 2 fs with SHAKE constraints on hydrogen bonds.

Dataset statistics:

- $\sim$ 1,400 diverse proteins from the PDB, spanning all major fold classes: α , β , α/β , IDPs.
- Proteins range from $\sim$ 50 to $\sim$ 500 residues.
- Aggregate simulation time: $\sim$ 1.2 μ s across all proteins and replicas.
- Train split: $\sim$ 1,265 proteins; Test split: $\sim$ 135 proteins.

During training, a random contiguous window of $T=16$ consecutive frames is extracted from each replica trajectory per minibatch. Overlapping windows are permitted as data augmentation.

7 Evaluation Protocol and Results

7.1 Evaluation Metrics

7.1.1 Root Mean Square Fluctuation (RMSF)

RMSF quantifies per-residue positional flexibility, measuring how much each residue's $\text{C}\alpha$ position deviates from its time-averaged position:

$$\text{RMSF}_i = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\mathbf{r}_i(t) - \langle \mathbf{r}_i \rangle\|^2} \quad (19)$$

High RMSF → flexible region (loops, termini). Low RMSF → rigid region (helices, β -sheets).

The primary summary metric is Pearson correlation r between generated and reference MD RMSF profiles, averaged over all test proteins.

7.1.2 Ramachandran Distribution: Jensen-Shannon Divergence

The Ramachandran plot captures the joint distribution of backbone dihedral angles (φ , ψ).

MDGen's generated (φ , ψ) distribution is compared to reference MD using the Jensen-Shannon divergence (JSD), discretised over a 36×36 grid. Lower JSD = better.

7.1.3 Temporal Metrics

Two metrics test temporal coherence—quantities impossible to evaluate from i.i.d. samplers:

- Frame-to-frame RMSD Δ_t : in real MD, consecutive frames differ by only ~ 0.1 – 0.3 Å C α RMSD. Independent samplers show arbitrarily large jumps.
- Temporal autocorrelation function $C(\tau) = \langle \delta \mathbf{r}(t) \cdot \delta \mathbf{r}(t+\tau) \rangle / \langle |\delta \mathbf{r}(t)|^2 \rangle$: measures structural persistence. Physical trajectories decay smoothly from $C(0)=1$. i.i.d. samplers have $C(\tau)=0$ for $\tau \geq 1$ by definition.

7.2 Quantitative Results

Method	RMSF r $\uparrow$	Ramach. JSD	Frame Δ RMSD	ACF Match	Speed
Reference MD	1.00	0.00	ground truth	ground truth	1 $\times$
MDGen (DDPM)	~ 0.87	~ 0.08	[Y] matches	[Y] matches	$\sim 10^4 \times$
MDGen (DDIM)	~ 0.85	~ 0.09	[Y] matches	[Y] matches	$\sim 10^5 \times$
Indep. frame diffusion	~ 0.71	~ 0.11	[N] rand. jumps	[N] zero lag	$\sim 10^4 \times$
Normal Mode Analysis	~ 0.64	~ 0.19	harmonic only	harmonic only	$\sim 10^3 \times$
Boltzmann generators	~ 0.68	~ 0.14	[N] i.i.d.	[N] zero lag	$\sim 10^3 \times$

MDGen achieves Pearson $r \approx 0.87$ on per-residue RMSF, substantially outperforming both NMA ($r \approx 0.64$) and independent-frame diffusion ($r \approx 0.71$). Improvement over independent-frame diffusion indicates that temporal context improves equilibrium ensemble quality as well as kinetics—temporal attention acts as an informative prior constraining physically accessible conformation space. Only MDGen reproduces correct temporal autocorrelation functions and frame-to-frame step-size distributions.

7.3 Generalisation to Out-of-Distribution Fast-Folding Proteins

MDGen is evaluated on miniproteins from long reference simulations that were NOT in the ATLAS training set: Trp-cage (20 residues), BBA motif (28 residues), villin headpiece (35 residues), WW domain (35 residues), NTL9 (39 residues), Chignolin (10 residues). MDGen correctly identifies qualitative flexibility profiles for these proteins—flexible C-terminal helices, rigid hydrophobic cores, elevated RMSF in loop regions—demonstrating generalisation across fold space beyond the ATLAS distribution.

8 Applications

8.1 Cryptic Pocket Discovery in Drug Design

Approximately 50% of protein drug targets undergo conformational changes critical for inhibitor binding. Cryptic pockets—absent in the static crystal structure but transiently open during dynamics (Oleinikovas et al., 2016)—are missed entirely by static docking, contributing to high false-negative rates in virtual screening.

MDGen workflow for cryptic pocket discovery:

- Generate 10^2 – 10^3 backbone trajectory frames using MDGen (seconds–minutes on one GPU).
- Run pocket detection software (FPOCKET, SiteMap) on each frame.
- Identify pockets that appear transiently across the ensemble.
- Prioritise pockets for downstream docking and lead optimisation.

This approach mirrors ensemble docking (Totrov & Abagyan, 2008) but replaces expensive multi- μ s MD with MDGen-generated ensembles 3–4 orders of magnitude faster. MDGen's correct temporal autocorrelation ensures conformational transitions consistent with natural protein dynamics.

8.2 Intrinsically Disordered Proteins (IDPs) and Disease

Intrinsically disordered proteins (IDPs; Uversky, 2019) lack a stable folded structure and exist as highly dynamic conformational ensembles. They are medically critical, frequently driving aberrant condensate formation and aggregation in neurodegenerative diseases.

- **α -Synuclein** (Parkinson's disease): ~140-residue IDP; conformational ensemble determines nucleation and growth pathways of toxic amyloid fibrils.
- **Tau** (Alzheimer's disease): highly flexible repeat domain; disordered regions initiate neurofibrillary tangle formation.
- **p53** (cancer): most frequently mutated tumour suppressor; disordered N/C-terminal regions mediate transactivation through conformational dynamics.

Static predictors such as AlphaFold2 report low-confidence (pLDDT < 50) predictions for disordered regions—a diagnostic flag, not a structural answer. MDGen generates backbone conformational ensembles for these regions, providing the distribution of conformations rather than a misleading single coordinate set.

8.3 Ensemble-Aware Protein Engineering

- Mutational scanning: rapidly generate conformational ensembles for point mutants to predict effects on dynamics without re-running MD.
- Allosteric pathway mapping: condition on two allosteric states and use interpolation mode to identify signal-transmitting residues.
- Antibody CDR loop flexibility: characterise CDR loop dynamics governing antigen recognition and binding affinity.

9 Limitations

- **Fixed timescale:** $T=16$ frames ~ 16 ns. Millisecond-scale processes (full folding, large domain translocations) are out of scope.
- **Training distribution bias:** The dataset predominantly contains single-chain, globular, soluble proteins (50–500 residues). Membrane proteins, large complexes, and heavily post-translationally modified proteins are under-represented.
- **Implicit solvents:** MDGen learns from explicit-solvent MD but does not model solvent explicitly. Hydration shell dynamics and specific ion binding are only implicitly captured insofar as they manifest in backbone motions.
- **Force-field dependency:** Trajectories emulate AMBER ff14SB statistics. Systematic force-field errors are inherited by MDGen—it is a data-driven emulator, not an independent physics engine.
- **Backbone-only representation:** side-chain coordinates must be reconstructed post-hoc using Rosetta Relax or DLPacker, limiting direct use in atom-level applications (e.g. binding free energy calculations).
- **Rare barrier-crossings:** conformational transitions requiring crossing high energy barriers (rare events on $\mu\text{s}+$ timescales) are statistically improbable within a 16-frame window.

10 Future Directions

- **Full-atom generation:** jointly model backbone frames and all-atom side-chain coordinates as a unified diffusion model—enabling direct computation of binding free energies and pharmacophoric features.
- **Longer timescales:** on μs -scale datasets (D.E. Shaw Research Anton) or enhanced-sampling trajectories (metadynamics, REST2) to unlock rarer conformational transitions.
- **Protein-ligand dynamics:** model receptor-ligand systems for ensemble-aware docking and entropic contributions to binding affinity ($\Delta G = \Delta H - T\Delta S$).
- **Experimental conditioning:** incorporating NMR chemical shifts, SAXS profiles, or cryo-EM density maps as soft conditioning signals during inference for physics-guided ensemble refinement.
- **Beyond proteins:** RNA, DNA, carbohydrates, and lipid membranes all undergo important conformational dynamics. The SE(3) trajectory diffusion framework can be adapted for these classes with appropriate geometric representations.

11 Conclusion

MDGen represents a conceptual advance in generative molecular modelling. By framing the problem as trajectory generation rather than independent equilibrium sampling, MDGen is the first model to simultaneously achieve:

- High per-residue RMSF correlation with reference MD ($r \approx 0.87$ vs NMA $r \approx 0.64$).
- Physically correct Ramachandran backbone dihedral distributions (JSD ≈ 0.08).
- Correct temporal autocorrelation functions and frame-to-frame step-size distributions — metrics that no prior independent-frame sampler can reproduce.
- $\sim 10,000\times$ speed-up over equivalent explicit-solvent MD simulation.

The architectural innovations enabling this — SE(3)-equivariant Invariant Point Attention (spatial) combined with temporal axial attention (kinetic) — provide a template for incorporating physical symmetries and temporal structure into generative models for other molecular and biological systems.

MDGen's central intellectual contribution: MD trajectories should be treated as the PRIMARY generative target, not a source of independent equilibrium samples. This reframing enables models to learn both the thermodynamics (what conformations are accessible) AND the dynamics (how fast they interconvert) of protein motion from data — a template for how generative AI can emulate complex physical processes.

References

- [1] Jing, B., Stärk, H., Jaakkola, T., & Berger, B. (2024). Generative Molecular Dynamics. arXiv:2409.17808. NeurIPS 2024. Code: github.com/bjing2016/mdgen
- [2] Vander Meersche, Y. et al. (2024). ATLAS: Protein Flexibility Description from Atomistic Molecular Dynamics Simulations. Nucleic Acids Research, 52(D1), D384–D392.
- [3] Jumper, J. et al. (2021). Highly accurate protein structure prediction with AlphaFold. Nature, 596(7873), 583–589.
- [4] Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. NeurIPS 2020, 33, 6840–6851.
- [5] Song, Y. et al. (2021). Score-based generative modeling through stochastic differential equations. ICLR 2021.
- [6] Song, J., Meng, C., & Ermon, S. (2021). Denoising diffusion implicit models. ICLR 2021.
- [7] Lin, Z. et al. (2023). Evolutionary-scale prediction of atomic-level protein structure with a language model. Science, 379(6637), 1123–1130.
- [8] Watson, J.L. et al. (2023). De novo design of protein structure and function with RFdiffusion. Nature, 620(7976), 1089–1100.
- [9] Adcock, S.A., & McCammon, J.A. (2006). Molecular dynamics: survey of methods. Chemical Reviews, 106(5), 1589–1615.
- [10] Shaw, D.E. et al. (2010). Atomic-level characterization of structural dynamics. Science, 330(6002), 341–346.
- [11] Lindorff-Larsen, K. et al. (2011). How fast-folding proteins fold. Science, 334(6055), 517–520.
- [12] De Bortoli, V. et al. (2022). Riemannian score-based generative modelling. NeurIPS 2022, 35, 2406–2422.
- [13] Yim, J. et al. (2023). SE(3) diffusion model for protein backbone generation. ICML 2023.
- [14] Jing, B. et al. (2023). EigenFold: Generative protein structure prediction with

- diffusion models. ICLR 2023 Workshop.
- [15] Jing, B., Berger, B., & Jaakkola, T. (2024). AlphaFold meets flow matching for generating protein ensembles. ICML 2024.
 - [16] Lewis, S.M. et al. (2025). Scalable emulation of protein equilibrium ensembles. *Science*, 388(6748).
 - [17] Noé, F. et al. (2019). Boltzmann generators: sampling equilibrium states. *Science*, 365(6457), eaaw1147.
 - [18] Tirion, M.M. (1996). Large amplitude elastic motions in proteins. *Physical Review Letters*, 77(9), 1905.
 - [19] Maier, J.A. et al. (2015). ff14SB: improving accuracy of protein parameters. *J. Chem. Theory Comput.*, 11(8), 3696–3713.
 - [20] Frauenfelder, H., Sligar, S.G., & Wolynes, P.G. (1991). Energy landscapes and motions of proteins. *Science*, 254(5038), 1598–1603.
 - [21] Fuchs, F.B. et al. (2020). SE(3)-Transformers: 3D roto-translation equivariant attention networks. *NeurIPS 2020*, 33, 1970–1981.
 - [22] Satorras, V.G. et al. (2021). E(n) equivariant graph neural networks. *ICML 2021*.
 - [23] Vincent, P. (2011). A connection between score matching and denoising autoencoders. *Neural Computation*, 23(7), 1661–1674.
 - [24] Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. *ICLR 2019*.
 - [25] Lugmayr, A. et al. (2022). Repaint: Inpainting using denoising diffusion probabilistic models. *CVPR 2022*.
 - [26] Jing, B. et al. (2022). Torsional diffusion for molecular conformer generation. *NeurIPS 2022*, 35, 24240–24253.
 - [27] Ho, J. et al. (2019). Axial attention in multidimensional transformers. *arXiv:1912.12180*.
 - [28] Oleinikovas, V. et al. (2016). Understanding cryptic pocket formation. *J. Am. Chem. Soc.*, 138(43), 14257–14263.
 - [29] Uversky, V.N. (2019). Intrinsically disordered proteins and their physics. *Frontiers in Physics*, 7, 10.
 - [30] Ingraham, J.B. et al. (2023). Illuminating protein space with RFdiffusion. *Nature*, 623(7989), 1070–1078.